Chapter 7:
Ethical Considerations

Scientific Ethics

- The role of ethics
- Fundamental principles of scientific ethics
- Ethics and scientific conduct
- Ethics and scientific writing

Ethics

- A branch of philosophy that addresses questions about morality, that is, concepts like good and bad, right and wrong, justice and injustice, and virtue and evil.
- Ethics are defined as **a set of moral obligations that define the principles of right and wrong conduct** in a community or profession, and which can be used by individuals to guide their choices and behaviours.
- Many professions and communities have formalized ethical codes to guide professionals in their field
  - The Hippocratic Oath: doctors should „do no harm“ to their patients.
  - Professional engineers code of ethics: „hold paramount the safety, health, and welfare of the public.”
  - First law of robotics: “A robot may not injure a human being.”
- Ideally, ethical principles become ingrained in everyday professional practices – part of the way a profession is practiced.
- A breach of ethics is considered a very serious offence – punishable within the profession and sometimes also by law.

Key concepts in ethics

- Ethical behavior describes a set of actions that abide by certain rules of :
  - **Responsibility** means accepting the potential costs, duties, and obligations of one’s decisions.
  - **Accountability** means being answerable to others for decisions made and actions taken.
  - **Liability** means accepting responsibility and accountability so individuals can recover damages done to them through breaches of responsibility.
  - **Due diligence** means investigating or exercising care to ensure individuals can examine or appeal how responsibility, accountability, and liability are applied.

Key goal in scientific ethics

- Maintaining “honesty and integrity” in all stages of scientific conduct.
  - includes all aspects of scientific activity, such as experimentation, testing, education, data collection, data analysis, data storage, data sharing, peer review, etc.
  - Also other activities that have a direct bearing on science, such as government funding or staffing of research teams.

Six Fundamental Ethical Principles for Scientific Research

Scientific honesty

Scientists should not commit scientific fraud by, for example, fabricating, “fudging,” trimming, “cooking,” destroying, or misrepresenting data.

Six Fundamental Ethical Principles for Scientific Research

Carefulness

Scientists should avoid careless errors and sloppiness in all aspects of scientific work.

Example Implementation Carefulness

- Academic peer review processes

Six Fundamental Ethical Principles for Scientific Research

Intellectual freedom

Scientists should be free to pursue new ideas and criticize old ones and conduct research on anything they find interesting.

Example Implementation Intellectual Freedom

Tenure / “Verbeamtung”

- an indefinite appointment that can be terminated only for cause or under extraordinary circumstances that provides sufficient protection for scientists to pursue their ideas
  - e.g., in Germany: salary, health cover, pension, rooms, assistants, budget, etc.

- Objective: to safeguard academic freedom for all who teach and conduct research in higher education.
- Provides conditions for scientitss to pursue research and innovation and draw evidence-based conclusions free from corporate or political pressure.
- Should faculty members possibly lose their positions because of their speech, publications, or research findings, they cannot properly fulfill their core responsibilities to advance and transmit knowledge.

- Not without controversy, e.g., the lectures of Prof Dr Bernd Lucke at the University of Hamburg in 2019
  - https://www.uni-hamburg.de/en/newsroom/im-fokus/2019/1022-fragen-antworten-lucke.html
  - https://www.tagesspiegel.de/wissen/massive-stoerung-bei-vorlesung-von-afd-gruender-uni-hamburg-sagt-bernd-lucke-schutz-zu/25125924.html

Six Fundamental Ethical Principles for Scientific Research

Openness

Whenever possible, scientists should share data, results, methods, theories, equipment, and so on; allow people to see their work; and be open to criticism.

Six Fundamental Ethical Principles for Scientific Research

Attribution of credit

Scientists should not plagiarise the work of other scientists. They should give credit where credit is due but not where it is not due.

Six Fundamental Ethical Principles for Scientific Research

Public responsibility

Scientists should report research in the public media when the research has an important and direct bearing on human happiness and when the research has been sufficiently validated by scientific peers.

Ethical clearance considerations

- An IRB evaluates
  1. the extent to which participation in a study
     - is voluntary,
     - does not exert physical or psychological stress, and
     - not cause other kinds of damage to participants
  2. whether participants must give consent regarding
     - how their data will be used
     - how their data will be reported
     - how the data will be protected in terms of anonymity or confidentiality
  3. whether participants have the right to withdraw from participation at any time.
  4. how data is stored and analysed
     - Involves ownership, storage and backup, privacy, confidentiality, access, and reuse.

Ethical considerations for scientific writing

- A subset of ethical issues in scientific conduct that relates only to the reporting of research
- Very important part of scientific ethics because it is typically only through reported research that an ethical issue is revealed
  - we typically cannot learn about data fabrication or amendment until those data are disclosed.
  - We cannot identify a lack of attribution of credit until an unnamed contributor sees it in writing

1. Plagiarism

- the wrongful appropriation, close imitation, or purloining and publication of another author’s language, thoughts, ideas, or expressions and their representation as one’s own work.
- the act of passing off someone else's work as your own, whether intentionally or unintentionally.
- The most common form of scientific misconduct.

Forms of plagiarism

- Intentional plagiarism
  - a writer knowingly lifts text directly from other authors' work without giving appropriate credit.
- Duplicate publication
  - an author submits for publication a previously published work as if it were original.
- Self-plagiarism
  - a writer copies large parts of an earlier manuscript word for word into a new manuscript.
  - can occur when individuals pursue large programs of research over many years on the same topic, so they are constantly building on their own work and in their own language.

Protecting against plagiarism

1. Always acknowledge the sources of and contributions to your ideas.
2. Enclose in quotation marks any passage of text that is directly taken from another author's work and acknowledge that author in an in-text citation.
3. Acknowledge every source you use in writing, whether you paraphrase it, summarise it, or quote it directly.
4. When paraphrasing or summarising other authors' work, reproduce the meaning of the original author's ideas or facts as closely as possible using your own words and sentence composition.
5. Do not copy sections of your previously published work into a new manuscript without citing the publication and using quotation marks.

2. Recognition of co-author contributions

- Concerns the **appropriate acknowledgement** (not too much or too little) of collaborators' substantial contributions to a piece of scholarly work.

- An ethical issue that appears frequently in scientific work because collaboration is the norm, not the exception.
  - Working alone means less productivity
  - Working alone means having to do every thing well
  - Collaboration means sharing workload, complementing skills, broadening the domain of interest

- Recognizing co-author contributions appropriately be difficult to deal with because the correct attribution of credit sounds easy but is hard to identify in practice.

- Making co-authorship decisions is important because on the one hand co-authorship confers credit to individuals for their contribution to academic tasks, which can have academic, social, and financial implications; but on the other hand, co-authorship also implies responsibility and accountability for published works.

Four ethical issues relating to co-authorship

- Coercion authorship
  - occurs when intimidation is used to gain authorship credit, such as when a senior person pressures a more junior person to include the senior person's name on a paper to which he or she has not contributed enough to qualify for authorship.
- Gift authorship
  - Occurs when individuals are given recognition as co-authors without having made substantial contributions, often for reasons like acknowledging friendship, gaining favour, or giving the paper more legitimacy by adding well-known senior researchers to the list of authors.

Four ethical issues relating to co-authorship

- Mutual support authorship
  - occurs when two or more authors (or author groups) agree to place their names on each other's papers to enhance their perceived productivity. The "authors" can count both publications towards their own list of papers, receive citations for both papers, and so forth.
- Ghost authorship
  - occurs when papers are written by people who are not included as authors or are not acknowledged. A typical form of ghost authorship involves using or hiring professional scientific writers, perhaps because the researchers feel they cannot write "well" or "scientifically."

Managing coauthorship

- Golden rule of publishing: Good papers built on good research.
  - You can contribute in either or both areas.

- Most important involved in the research process warrant co-author recognition when done by someone else but not all of them!
  - Developing an original idea
  - Designing a study
  - Organizing data collection
  - Collecting data
  - Analyzing data
  - Writing and revising a paper

  - Sponsoring/funding the project
  - Managing the project

My decision rules

- A co-author has…
  1. made substantial contributions to the conception or design of the research or the acquisition, analysis, or interpretation of data for the research; and
  2. made substantial contributions to drafting the publication or revising it critically for important intellectual content; and
  3. given final approval of the version to be published; and
  4. agreed to be accountable for all aspects of the work, including being accountable for the parts of the work he or she has done, being able to identify which co-authors are responsible for other parts of the work, and having integrity about the contributions of other co-authors.

- Authorship of a research output should <u>not be claimed</u> when
  1. participation rests solely in the acquisition of funding or the collection of data.
  2. General supervision of the research group does not justify authorship.

- **These are my criteria; they may or may not be yours**. Rules change by country, institution, and sometimes persons.

- E.g., the DFG (“Deutsche Forschungsgemeinschaft”) demands “participation in” rather then “contribution to” the points above.

Authorship credit: what is the order of authors?

- The order of authors is entirely up to the authors.
- Typical practices when it comes to putting the authors' names on the paper…
  - Most often, the first authors are the ones who did most of the work, with the authors listed in descending order of contributions
  - Some put principal investigators at the end of the list
  - Some groups do it alphabetically
  - In some scientific fields the most highly credited author is the one whose name appears [first/last].
  - Social scientists tend to place the authors' names in alphabetical order regardless of the amount of effort that was contributed.
  - Some journals allow annotations identifying one or two authors who did the majority of the work (this can be important for PhD thesis defenses or for job applications)

A formal way of handling co-auhtor recognition: CRediT

- CRediT (Contributor Roles Taxonomy)
- A system introduced with the intention of recognizing individual author contributions, reducing authorship disputes and facilitating collaboration.
- Defines a set of roles that individuals can occupy in a research process
- CRediT statements can be included in submissions or in final papers
- Example CRediT statement
  - Zhang San: Conceptualization, Methodology, Software Priya Singh.: Data curation, Writing- Original draft preparation. Wang Wu: Visualization, Investigation. Jan Jansen: Supervision.: Ajay Kumar: Software, Validation.: Sun Qi: Writing- Reviewing and Editing.

CRediT roles

Term	Definition
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims
Methodology	Development or design of methodology; creation of models
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code
Validation	Verification, whether as a part of the activity or separate, of the overall replication/ reproducibility of results/experiments and other research outputs
Formal analysis	Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection
Resources	Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools
Data Curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later reuse
Writing - Original Draft	Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation)
Writing - Review & Editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre-or postpublication stages
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/ data presentation
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team
Project administration	Management and coordination responsibility for the research activity planning and execution
Funding acquisition	Acquisition of the financial support for the project leading to this publication

3. Honest reporting

- An ethical standard that demands that research publications comply with expectations for transparency, openness, and reproducibility.
- Typical ethical issues
  - Publication bias
    - systematic suppression of a certain type of research results in published papers, such as negative hypothesis tests or replications, or
    - systematic preference given to innovative and novel findings rather than confirmations of known findings
  - HARKing (hypothesizing after the results are known)
  - *p*-Hacking (the misuse of data analysis to find patterns in data that can be presented as statistically significant)

Publication bias

- occurs when the outcome of an experiment or research study influences the decision whether to publish or otherwise distribute it.
- Known as the file-drawer problem: often investigators decline to submit results when they are found not to support initial hypotheses
- Consequence: the publication of “negative” or “insignificant” results is impeded
- Then, published studies are no longer a representative sample of the available evidence.

HARKing

- The false portrayal of a post hoc hypothesis as if it were an a priori hypothesis.
  - Can invalidates the idea of a priori hypothesis generation and subsequent testing.
  - Can lead to scholars not communicating valuable information about what did not work
  - Can lead to distorted publications limited to ideas and findings without a faithful representation of the scientific process through which these ideas were born.
  - Risks increasing levels of Type 1 errors: if one attempts (too) many post hoc analyses on the same data, some tests will generate false positives simply by chance
  - Risks favoring weaker theories that post hoc accommodate results rather than correctly predict them.

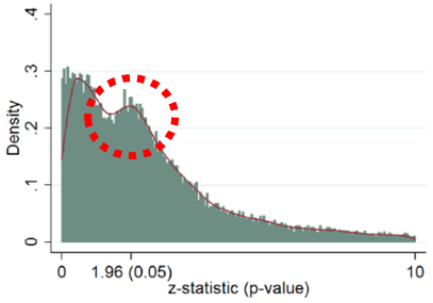
p-Hacking

- occurs when researchers collect or select data or statistical analyses until nonsignificant results become significant.
- Significant results increase the chance of being published but when published data are biased, data synthesis might lead to flawed conclusions.
- Means that we do not know if the strength of the relationship found is purely an artifact of the sample, the analytical method used, or legitimate judgment calls made by the researcher.

Economics

Brodeur et al (AEJ:A, in press)

"Star Wars: The empirics strike back"

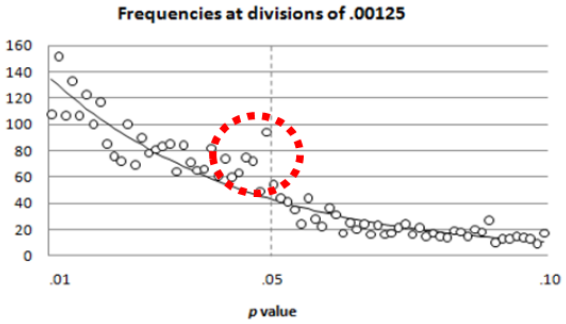


(b) De-rounded distribution of z-statistics.

Psychology

Masicampo Lalande (QJEP, 2012)

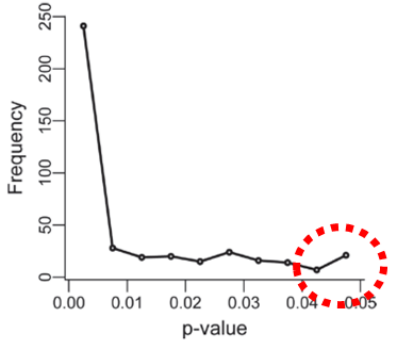
"A peculiar prevalence of p values just below .05"



Biology

Head et al (PLOS Biology 2015)

"Extent and Consequences of P-Hacking in Science"



Recommendations for honest reporting

- **Pre-registration**: make hypotheses public prior to data colletion and analysis
- **Open science**: publicly share raw data used in analysis
- **Register procedures**: Make analyses (e.g., codes, programs) available to others
- **Conduct replications**: repeat studies to see if results remain robust

4. Appropriate use of language

- Refers to the wording of scientific reports so they are not biased in terms of gender, race, orientation, culture, or any other characteristics.
- Stipulates using gender-responsible, ethnicity-responsible, and inclusive language wherever possible.
- Guidelines:
  - **Specificity**
    - describe specific behaviours rather than stereotypes: e.g., calling a behavior “dominant and opinionated” instead of “typically male”.
  - **Labelling**
    - Refer to concrete labels rather than abstract class tags, e.g., referring to countries’ populations—Mexicans or Chinese—instead of classes like “Hispanics” or “Asians”
  - **Professional acknowledgments**
    - Use professional classifications, not personal labels, like “medical practitioner” or “doctor” instead of “female doctor.”

Summary: Ethics in information systems research

- The role of ethics:
  - Professional code of conducts for IS scientists
- Fundamental principles of scientific ethics:
  - Maintaining honesty and integrity through the scientific process
- Ethics and scientific conduct:
  - Ethical clearance and independent ethics review
- Ethics and scientific writing:
  - Plagiarism, publication bias, honest reporting and appropriate language.

End of Chapter 7

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